Sadeeq Adeyemi

Perception Engineer, 3D Computer Vision and Multi-Object Tracking

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SUMMARY

Perception engineer in 3D computer vision and deep learning. I build trackers that keep an object's identity while it is hidden, and prove them on public benchmarks and real driving data. Shipping machine learning in industry since 2018, from a vision model on a steel plant floor to research on Meta's SAM 3.

SKILLS

Tracking and estimation: multi-object tracking and segmentation (MOTS), Kalman filters (EKF, UKF), data association (Hungarian, JPDA, MHT), sensor fusion

3D vision and deep learning: lidar point clouds, camera and lidar projection, ray casting, instance segmentation, SAM 3, YOLOv8, re-identification

Software: Python, C++17, PyTorch, ROS 2, OpenCV, filterpy, MATLAB, Git, Docker, Linux, AWS, AI coding agents

Evaluation: HOTA, IDF1 and MOTA with TrackEval, significance testing; KITTI, KITTI-MOTS, MOTS20, MOT17, NVIDIA PhysicalAI-AV

EXPERIENCE

Graduate Research Assistant, CVAR Lab (Computer Vision and Robotics), University of Maine 2025 – present

- Benchmarked Meta's SAM 3 and SAM 3.1 on KITTI-MOTS and MOTS20 with the official HOTA code on A100 GPUs: SAM 3 scored 82.6 car HOTA (val) and kept identity through 94% of 245 real occlusions; SAM 3.1, 74.6 and 63%.
- Traced about half of SAM 3.1's deficit to its shipped tracking settings by reading Meta's source code; a controlled rerun raised its car HOTA from 74.0 to 78.3. Isolated a separate fault that halts it on long, crowded sequences.
- Built a Kalman-filter identity layer on SAM's output that returns a lost ID when the object reappears, raising SAM 3.1's identity kept through occlusion from 63% to 70% ($p = 0.004$).
- Built a pipeline that mined 10,871 pedestrian occlusions from 1,256 NVIDIA driving clips in 12 countries, using 3D ray casting against labelled boxes and lidar depth projected into a fisheye camera.
- Showed on that real street geometry that an occlusion-aware tracker keeps identity through 1 to 5 second hides in 57% of 1,600 cases, where a standard tracker keeps 2%; driven by a real YOLOv8 detector, 44% against 3%.
- Built mobile robots end to end, from sensors and wiring to ROS 2 nodes; implemented EKF localization fusing IMU, wheel odometry and lidar, with costmap planning and obstacle avoidance.

Materials Engineer, R.W. Gillespie, Maine 2024

- Ran cone penetration tests (CPT) and materials testing for construction projects in Maine.

Machine Learning Engineer and Structural Steel Fabricator, NewTech Steel, Nigeria 2018 – 2023

- Built and deployed a computer vision model that detects fatigue and defects in steel in real time, halving inspection time by automating a manual test.
- Led fabrication and erection of structural steel for factory and auditorium projects.

Civil Engineering Intern, Federal Capital Development Authority, Abuja, Nigeria Feb – Aug 2017

- Worked on the construction of wastewater treatment facilities for Nigeria's Federal Capital Territory.

PROJECTS

Occlusion-Aware 3D Multi-Object Tracker, C++17, github.com/sadeeqgandalf/occlusion-tracker 2026

- Kept identities better than the AB3DMOT baseline on KITTI val (association accuracy 77.7 vs 76.6 cars, 49.4 vs 48.4 pedestrians) at 0.15 ms per frame, by modelling track existence from measured detection rates and sight lines.

EDUCATION

MS, Data Science and Engineering, University of Maine 2025 – present

MS, Structural Engineering, University of Maine 2023 – 2025

MSc, Project Management, University of Portsmouth, United Kingdom 2023

BEng, Civil Engineering, Federal University of Technology Akure, Nigeria 2019

Certifications: AWS Certified Machine Learning Specialty; AWS Certified Solutions Architect, Associate